instances.

A good example of such a statistical law is the dependence of the magnitude of gas pressure on the density and temperature. The pressure of a gas is produced by the continual impact against the walls of the vessel of a vast number of gas molecules moving at random in all directions with great velocity. A summary computation of the aggregate dynamic effect of their impact reveals that the pressure against the walls of the vessel is almost proportional to the density of the gas as well as to the mean square of the molecular velocity, a result which is in satisfactory agreement with measurement if we regard the temperature as a measure of molecular velocity.

A direct confirmation of this theory of gas pressure is furnished by investigations on the temporal variations of the pressure against a very small portion of the wall of the vessel. Such variations, produced by random molecular impacts, can be observed wherever molecules in rapid flight come in contact with easily movable bodies; they manifest themselves also in the Brownian molecular movement, as well as in the fact that very sensitive scales never come to a complete rest, but constantly execute minute irregular vibrations about their position of equilibrium.

Analogously with the gas laws, the indeterminists attribute every other kind of physical regularity, ultimately to the operation of chance. They see nature ruled exclusively by statistics, and their aim is to base all physics on the calculus of probabilities

Actually, physical science has developed up to now on the very opposite foundation. It chose the *second* one of the two alternatives mentioned above: In order to be able to maintain the full and absolute validity of the law of causality, it modified slightly the basic proposition, that an occurrence is causally determined if it can be predicted with a certainty. This was done by using the word “occurrence” in a somewhat modified sense. Thus, theoretical physics considers as an occurrence not an actual individual process of measurement—which always includes accidental and unessential elements, too—but a certain, merely theoretical process; and in this manner it replaces the sense world, as given to us directly by our sense organs (or alternatively, by the measuring instruments which serve us as sharpened sense organs), by another world, the world picture of physics, which is a conceptual structure, arbitrary to a certain degree, created for the purpose of getting away from the uncertainty involved in every individual measurement and for making possible a precise interrelation of concepts.

Consequently, every measurable physical magnitude, every length, every time interval, every mass, every charge, has a double meaning, according as we regard it as directly given by some measurement, or conceive of it as translated into the world picture of physics. In the first interpretation, it is never capable of a sharp definition, and can therefore never be represented by